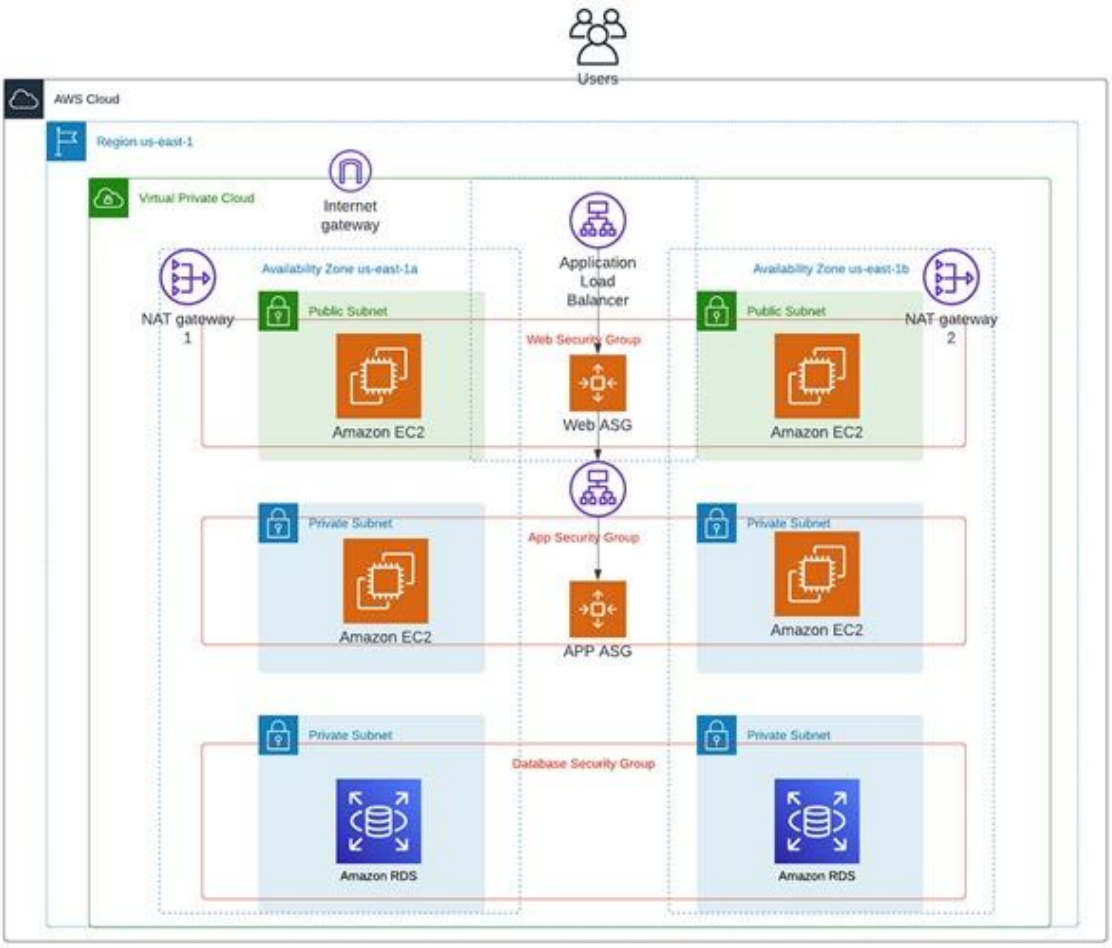
Name: S. Saran Kumar

PROJECT – 1

3-TIER ARCHITECTURE ON AMAZON CLOUD



Introduction: A three-tier application divides a system into three logical layers. The presentation tier handles the user interface, the application tier processes requests and business rules, and the data tier stores information. Separating these responsibilities makes it easier to update one layer without redesigning the entire system.

AWS provides networking, compute, load balancing, monitoring, and managed database services that can be combined to build this architecture. This project uses an Amazon VPC with public and private subnets, an Application Load Balancer (ALB), Amazon EC2 application instances, and Amazon RDS.

Architecture overview:

3 -tier architecture which the application is divided into three layers.

1. Presentation Layer: user interface where user I can interact with the system.

2. Application Layer : process requests and Coordinate between UI and database.

3. Data Base Layer: stores and manages application data.

AWS Services Used:

Amazon VPC: Creates an isolated virtual network and controls subnet routing.

Public and private subnets: Separate internet-facing components from application and database resources.

Internet Gateway: Enables internet routing for resources in public subnets when routes and security rules permit it.

Application Load Balancer: Receives HTTP/HTTPS traffic and routes requests to healthy targets.

Amazon EC2: Runs the web/application code.

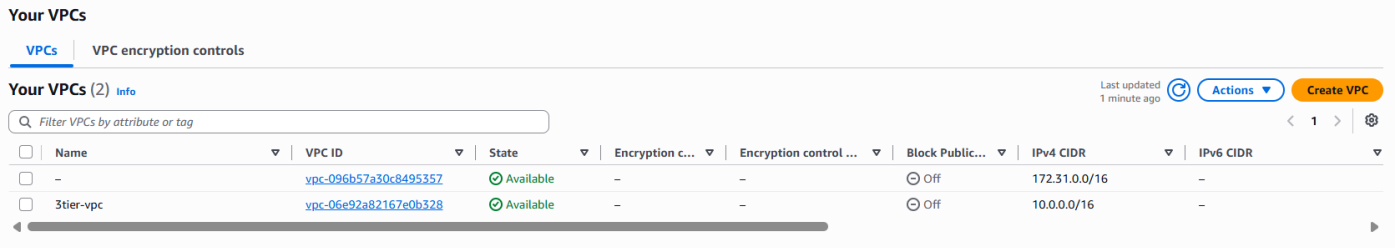
An Auto Scaling group can replace unhealthy instances and adjust capacity.

Amazon RDS: Provides a managed relational database such as MySQL or PostgreSQL.

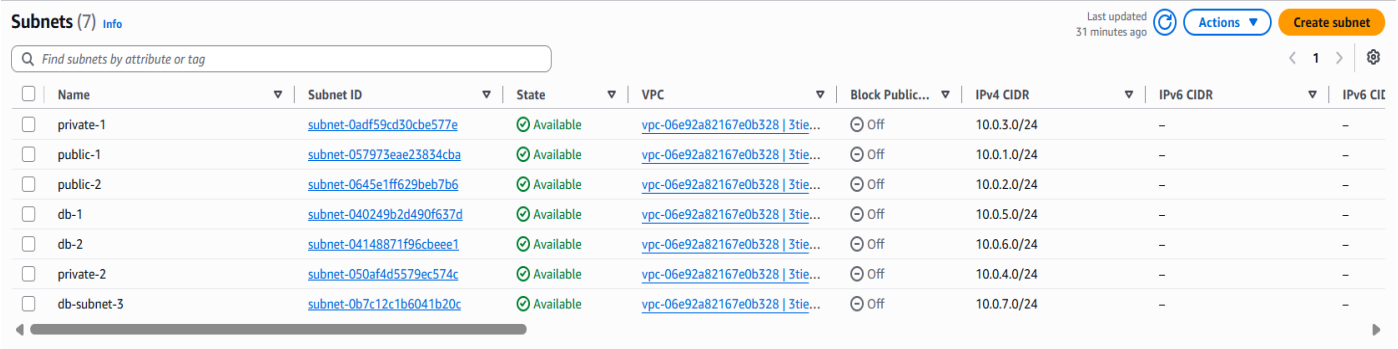
Database access is restricted to the application tier. Security Groups: Stateful virtual firewalls attached to resources.

Rules should allow only the required source, protocol, and port. Steps Involved in Creation Of 3-Tier Web Application on AWS:

1) Create VPC (Virtual Private Cloud) in any region.



2) Create two public subnets and four private subnets.



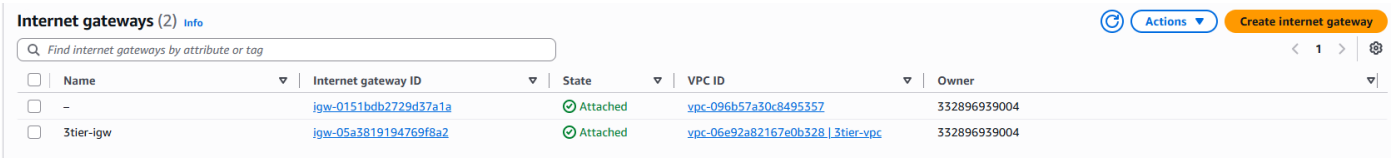
Subnets (7) Info

Last updated 31 minutes ago | Actions | Create subnet

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIE
private-1	subnet-0adf59cd30cbe577e	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.3.0/24	–	–
public-1	subnet-057973eae23834cba	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.1.0/24	–	–
public-2	subnet-0645e1ff629beb7b6	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.2.0/24	–	–
db-1	subnet-040249b2d490f637d	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.5.0/24	–	–
db-2	subnet-04148871f96cbeee1	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.6.0/24	–	–
private-2	subnet-050af4d5579ec574c	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.4.0/24	–	–
db-subnet-3	subnet-0b7c12c1b6041b20c	Available	vpc-06e92a82167e0b328 | 3tie...	Off	10.0.7.0/24	–	–

3) Create internet gateway and attach internet gateway to public subnets.



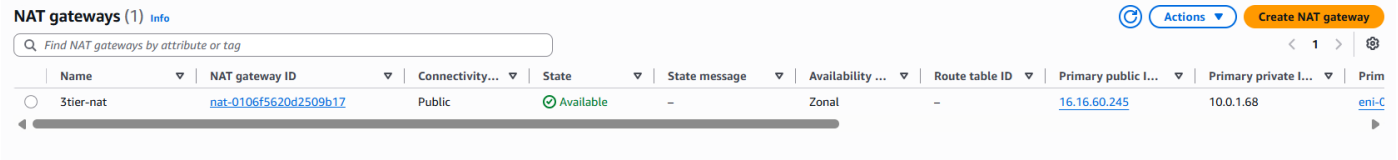
Internet gateways (2) Info

Actions | Create internet gateway

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
–	igw-0151bdb2729d37a1a	Attached	vpc-096b57a30c8495357	332896939004
3tier-igw	igw-05a3819194769f8a2	Attached	vpc-06e92a82167e0b328 | 3tier-vpc	332896939004

4) Create NAT gateway.



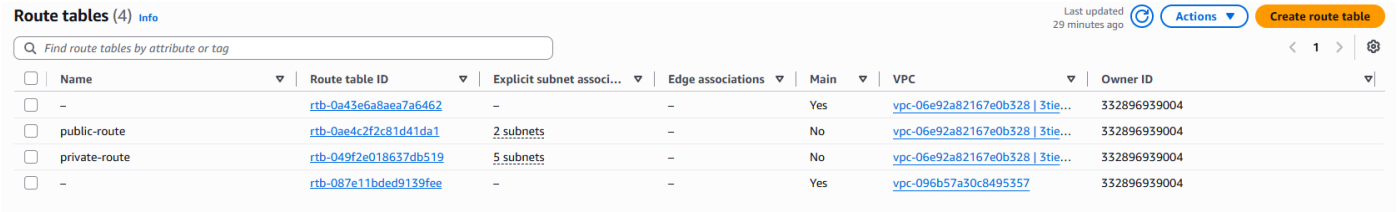
NAT gateways (1) Info

Actions | Create NAT gateway

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	Primary public I...	Primary private I...	Prim
3tier-nat	nat-0106f5620d2509b17	Public	Available	–	Zonal	–	16.16.60.245	10.0.1.68	eni-0

5) Create public route table and private route table and edit the routes.



Route tables (4) Info

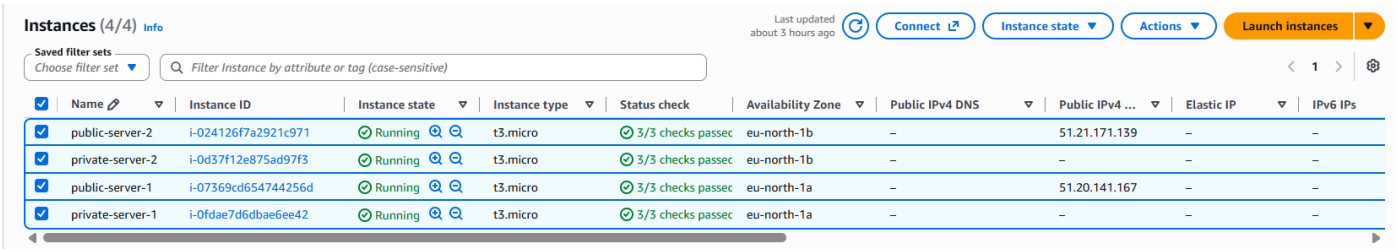
Last updated 29 minutes ago | Actions | Create route table

Find route tables by attribute or tag

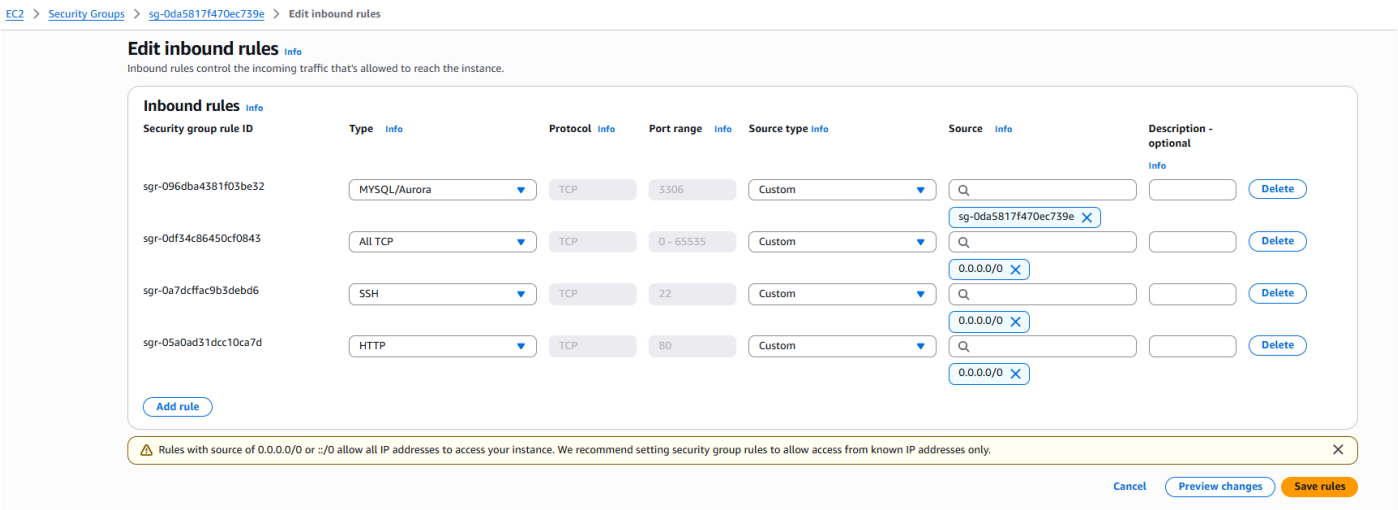
Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
–	rtb-0a43e6a8aea7a6462	–	–	Yes	vpc-06e92a82167e0b328 | 3tie...	332896939004
public-route	rtb-0ae4c2f2c81d41da1	2 subnets	–	No	vpc-06e92a82167e0b328 | 3tie...	332896939004
private-route	rtb-049f2e018637db519	5 subnets	–	No	vpc-06e92a82167e0b328 | 3tie...	332896939004
–	rtb-087e11bded9139fee	–	–	Yes	vpc-096b57a30c8495357	332896939004

Attach Internet Gateway to the Public route table and NAT Gateway to the Private Nat Gateway through Edit Routes.

6) Creation of Ec2 instances with ubuntu linux.



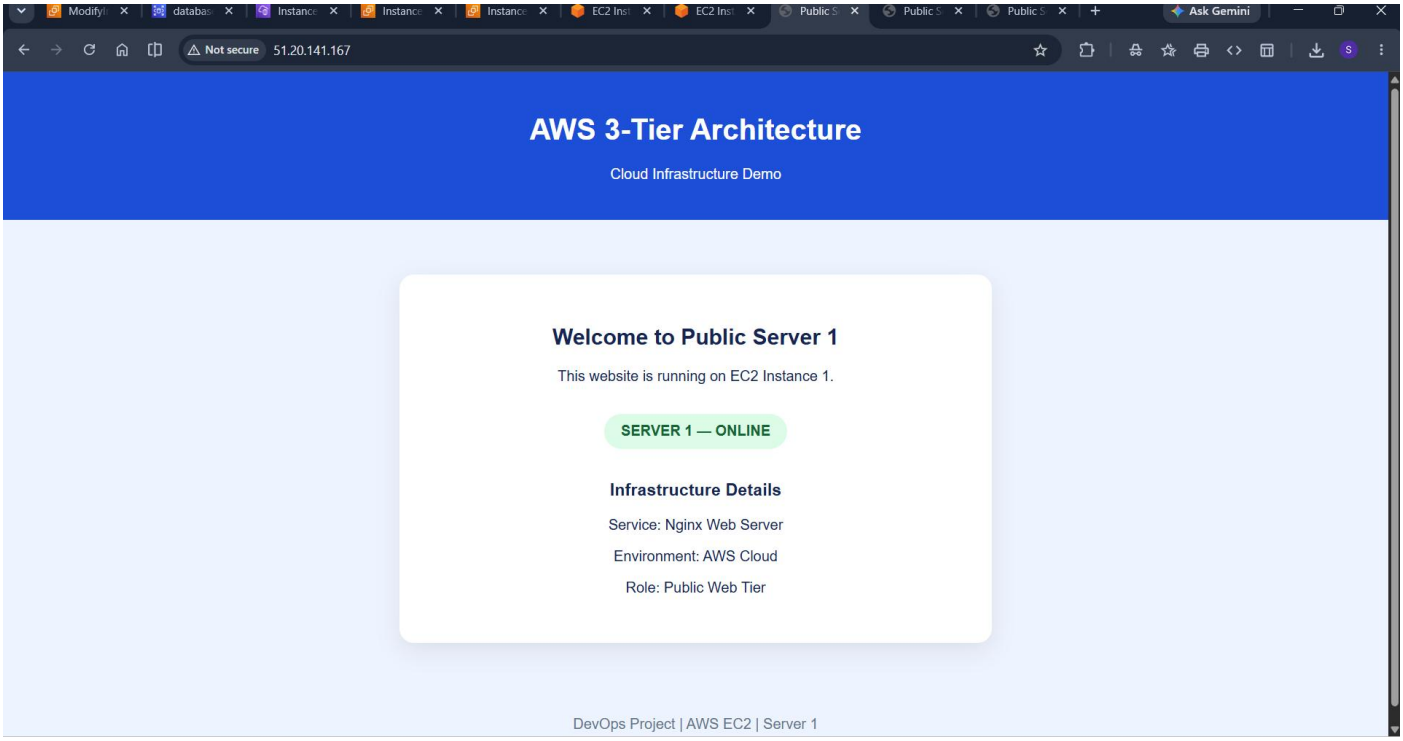
7) Create the security group and edit the inbound rules.



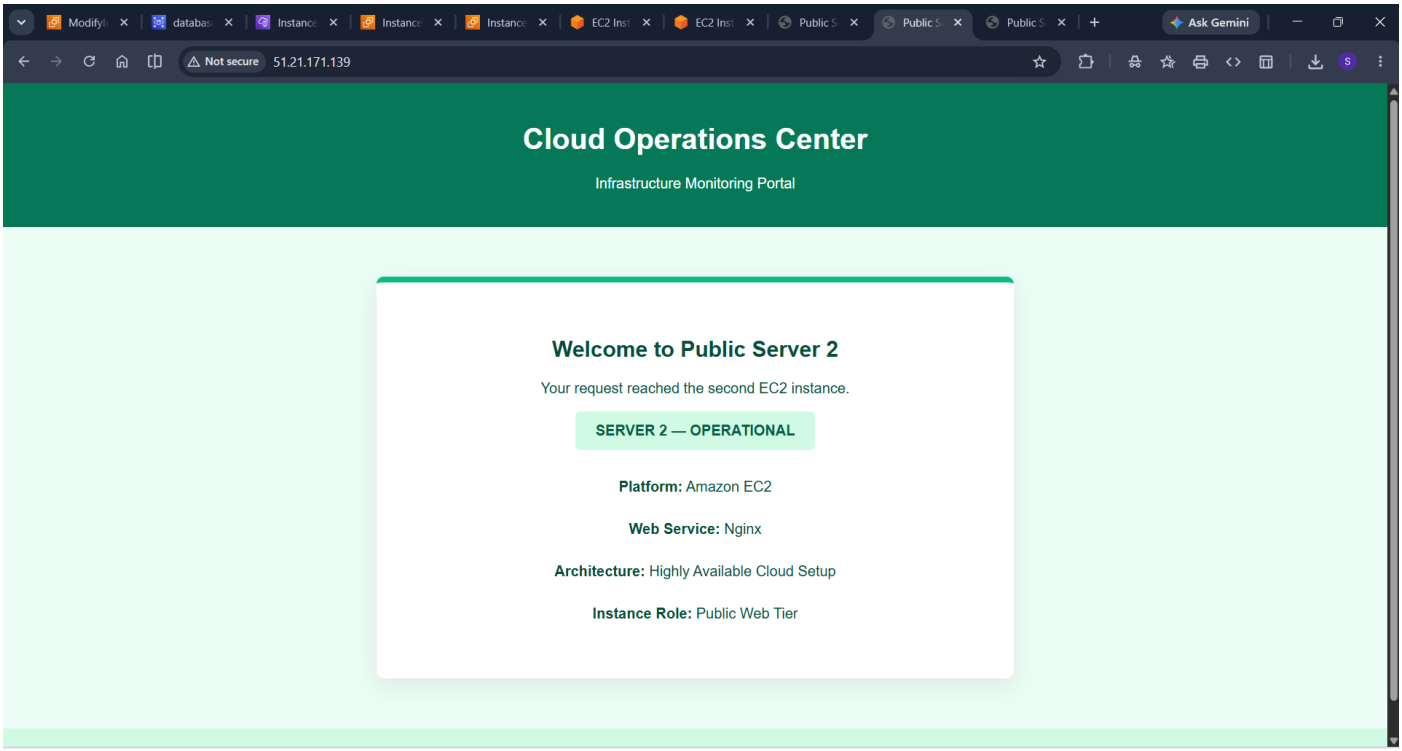
8) Connect the both public instances and install nginx server.

After installing remove the nginx index.html and create your own index.html.

PUBLIC SERVER 1:

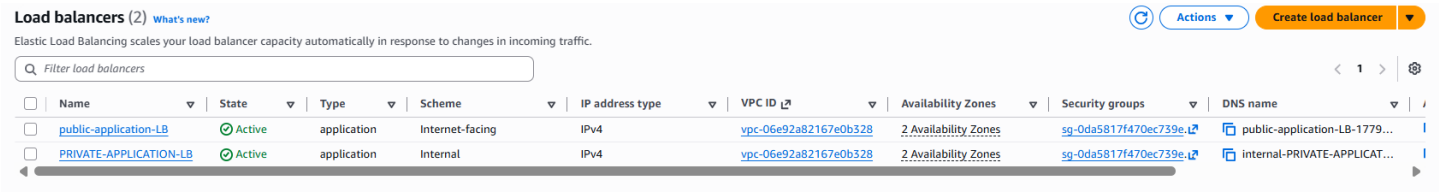


PUBLIC SERVER 2:



10) Create the target groups and attach to load balancer

*) Creation of load balancer.



Load balancers (2) What's new?

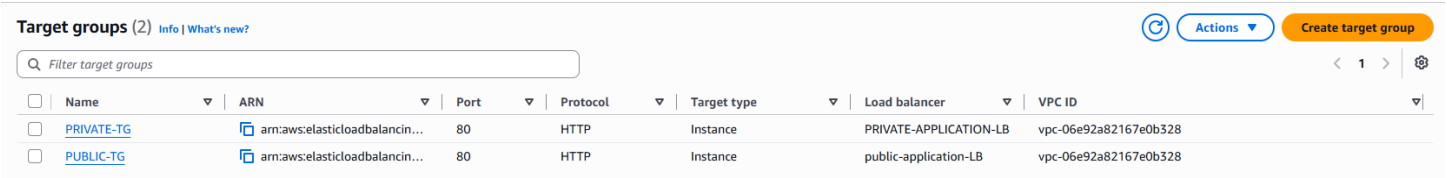
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Actions | Create load balancer

Filter load balancers

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
public-application-LB	Active	application	Internet-facing	IPv4	vpc-06e92a82167e0b328	2 Availability Zones	sg-0da5817f470ec739e	public-application-LB-1779...
PRIVATE-APPLICATION-LB	Active	application	Internal	IPv4	vpc-06e92a82167e0b328	2 Availability Zones	sg-0da5817f470ec739e	internal-PRIVATE-APPLICAT...

*) creation of Target groups and attach target group to load balancer



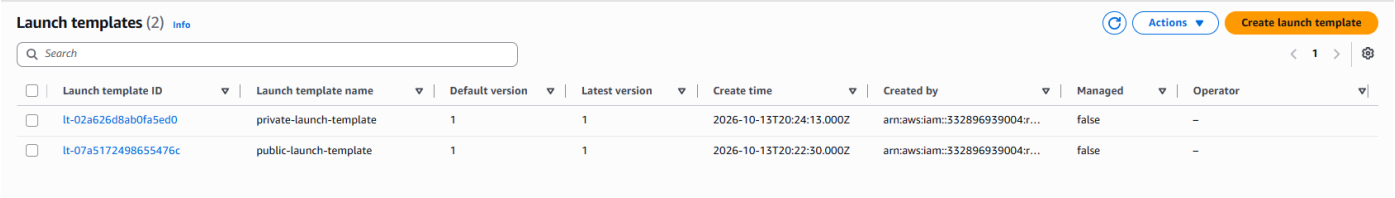
Target groups (2) Info | What's new?

Actions | Create target group

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
PRIVATE-TG	arn:aws:elasticloadbalancin...	80	HTTP	Instance	PRIVATE-APPLICATION-LB	vpc-06e92a82167e0b328
PUBLIC-TG	arn:aws:elasticloadbalancin...	80	HTTP	Instance	public-application-LB	vpc-06e92a82167e0b328

11) Create a template by using ec2 instance and use it in auto scaling.



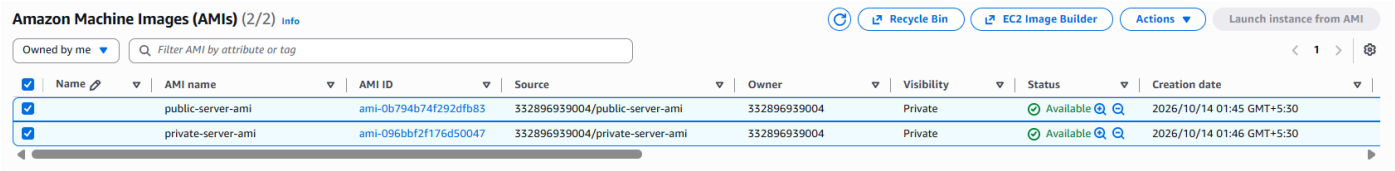
Launch templates (2) Info

Actions | Create launch template

Search

Launch template ID	Launch template name	Default version	Latest version	Create time	Created by	Managed	Operator
lt-02a626d8ab0fa5ed0	private-launch-template	1	1	2026-10-13T20:24:13.000Z	arn:aws:iam::332896939004:r...	false	–
lt-07a5172498655476c	public-launch-template	1	1	2026-10-13T20:22:30.000Z	arn:aws:iam::332896939004:r...	false	–

12) Creation of AMI images.



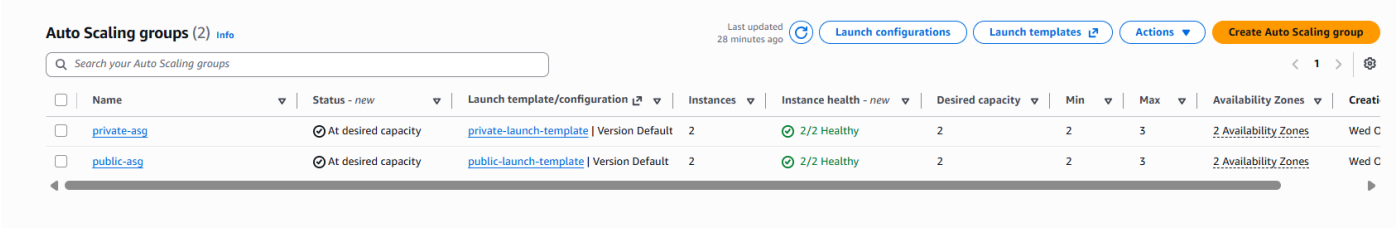
Amazon Machine Images (AMIs) (2/2) Info

Recycle Bin | EC2 Image Builder | Actions | Launch instance from AMI

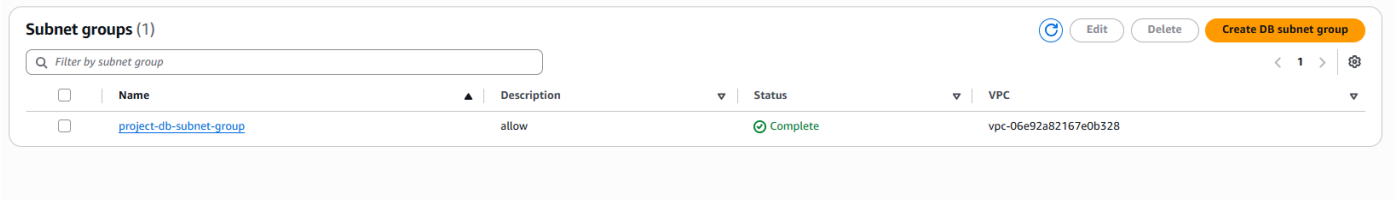
Owned by me | Filter AMI by attribute or tag

Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date
	public-server-ami	ami-0b794b74f292dfb83	332896939004/public-server-ami	332896939004	Private	Available	2026/10/14 01:45 GMT+5:30
	private-server-ami	ami-096bbf2f176d50047	332896939004/private-server-ami	332896939004	Private	Available	2026/10/14 01:46 GMT+5:30

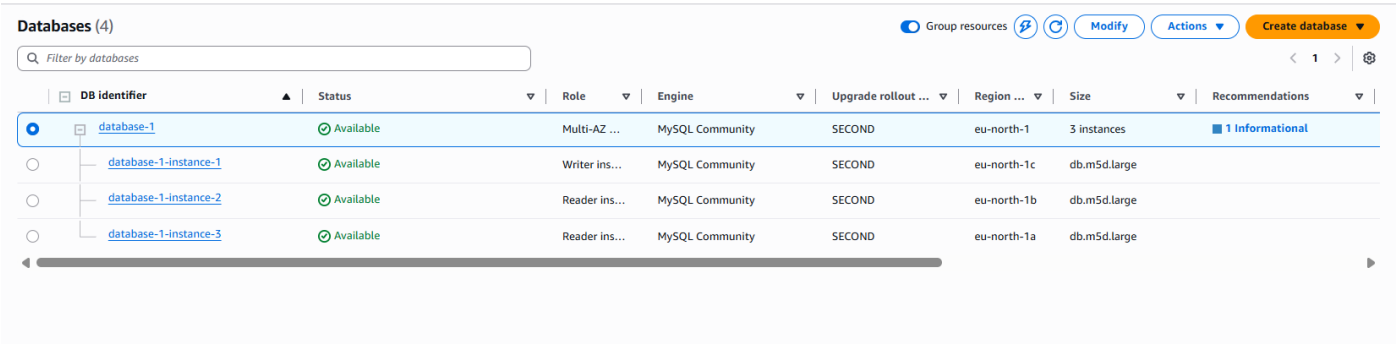
13) Creation of Auto Scaling groups and configure properly.



14) Create Subnets Group.



15) Create a database by using database cluster



16) Connect to private instance and install mysql in it.

*) Commands used to install sql

1. sudo apt install mysql-server.
2. sudo systemctl start mysql.service.

*) Connect to the created database by using the following command.

mysql -h endpoint url -u admin -p

17) create table and insert data into the table in the existing database by using the Writer endpoint.

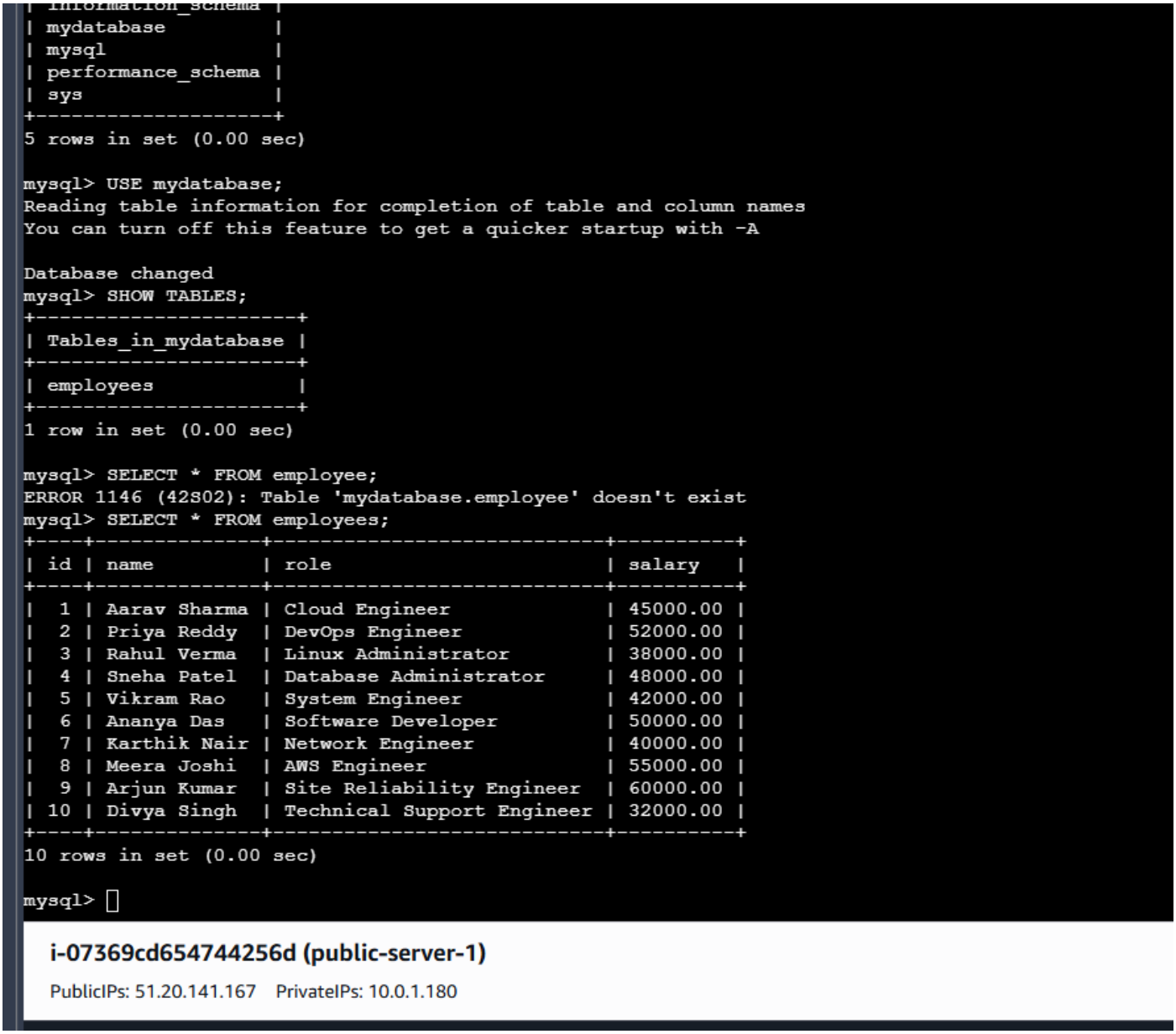
```
mysql> SELECT * FROM employees;
Empty set (0.00 sec)

mysql> INSERT INTO employees (name, role, salary)
    -> VALUES
    -> ('Aarav Sharma', 'Cloud Engineer', 45000.00),
    -> ('Priya Reddy', 'DevOps Engineer', 52000.00),
    -> ('Rahul Verma', 'Linux Administrator', 38000.00),
    -> ('Sneha Patel', 'Database Administrator', 48000.00),
    -> ('Vikram Rao', 'System Engineer', 42000.00),
    -> ('Ananya Das', 'Software Developer', 50000.00),
    -> ('Karthik Nair', 'Network Engineer', 40000.00),
    -> ('Meera Joshi', 'AWS Engineer', 55000.00),
    -> ('Arjun Kumar', 'Site Reliability Engineer', 60000.00),
    -> ('Divya Singh', 'Technical Support Engineer', 32000.00);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM employees;
+----+--------------+---------------------------+----------+
| id | name         | role                      | salary   |
+----+--------------+---------------------------+----------+
|  1 | Aarav Sharma | Cloud Engineer            | 45000.00 |
|  2 | Priya Reddy  | DevOps Engineer           | 52000.00 |
|  3 | Rahul Verma  | Linux Administrator       | 38000.00 |
|  4 | Sneha Patel  | Database Administrator    | 48000.00 |
|  5 | Vikram Rao   | System Engineer           | 42000.00 |
|  6 | Ananya Das   | Software Developer        | 50000.00 |
|  7 | Karthik Nair | Network Engineer          | 40000.00 |
|  8 | Meera Joshi  | AWS Engineer              | 55000.00 |
|  9 | Arjun Kumar  | Site Reliability Engineer | 60000.00 |
| 10 | Divya Singh  | Technical Support Engineer| 32000.00 |
+----+--------------+---------------------------+----------+
10 rows in set (0.00 sec)

mysql> EXIT;
Bye
ubuntu@ip-10-0-3-124:~$ hostname -I
10.0.3.124
ubuntu@ip-10-0-3-124:~$
```

18) Connect to public instance and download sql in it and connect to Reader end point and see the table created in the database.



19) Conclusion:

A three-tier web application on AWS separates the public entry point, application processing, and persistent data. With an Application Load Balancer, EC2 Auto Scaling, private subnets, Amazon RDS the design provides a practical foundation for security, monitoring, and growth. The architecture should be tested and adjusted to the application's requirements before production use, with particular attention to access controls, backups, resilience, and cost.